Rigid Body Dynamics

DISCLAIMER

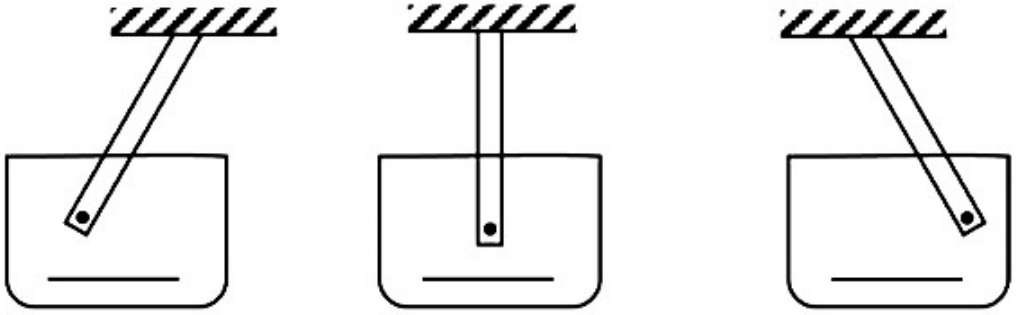
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Rigid Body Motion

Only Rigid Bodies Can Perform Rotation.
A body which is a collection of a large number of particles, moving in a constrained manner is termed as rigid body. The separation between any two particles on a rigid body remains constant. This is known as rigid body constraint.

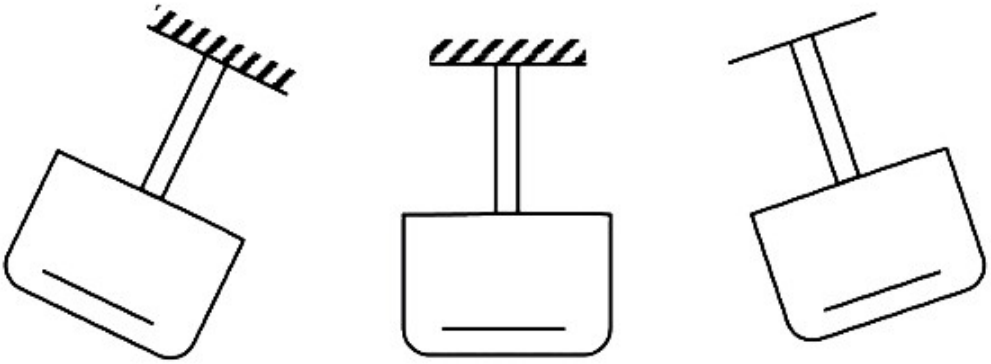
Different types of motion of a Rigid Body:

Translation: A rigid body is said to be in pure translation if any line drawn on it remains parallel to itself during complete motion.



In pure translation, all the particles of the body have equal velocity and acceleration at all instants and they cover equal distance and displacement in equal time.

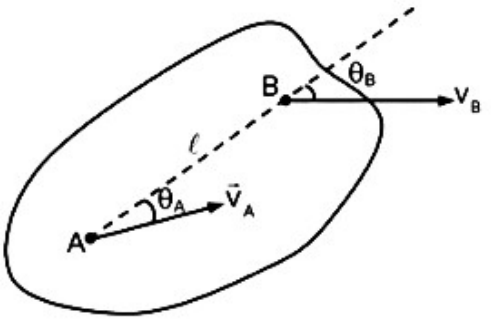
Rotation: A rigid body is said to be in rotation if any line drawn on it does not remain parallel to itself during complete motion. e.g.: A ceiling fan or wheel of a moving vehicle.



Rotation about a Fixed Axis: When at least one point of the body remains fixed, then motion of the rigid body is said to be rotation about a fixed axis.
For a 2D body, there will be one point that remains at rest (for disc or plate).
For a 3D rigid body, there will be a set of points that remain at rest.

Rigid Body Constraint

Consider a rigid body of an arbitrary shape as shown in the figure. Lets say that any two points A and B on the rigid body have velocities $\vec{v}_A$ and $\vec{v}_B$ as shown in the figure.



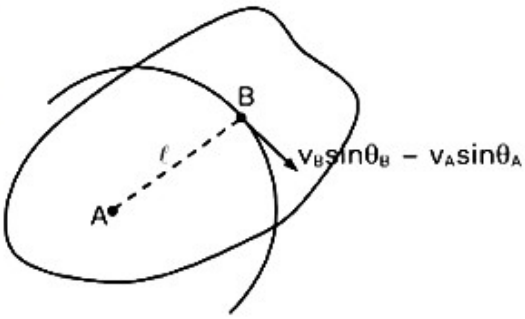
Now, as per rigid body constraint, the distance AB will be constant. Then, relative velocity of A wrt B or B wrt A along the line joining A and B should be zero.

$\therefore \quad v_A\cos\theta_A - v_B\cos\theta_B = 0$

$\Rightarrow \quad v_A\cos\theta_A = v_B\cos\theta_B$

Note that the components of velocities perpendicular to the line joining the points A and B may or may not be equal.

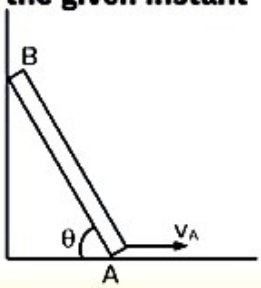
If the perpendicular velocity components are not equal, i.e., $v_A\sin\theta_A \neq v_B\sin\theta_B$, then from the frame of reference of point A, point B is in circular motion with A as centre.



Angular velocity of the line AB,

$$\omega = \frac{v_B\sin\theta_B - v_A\sin\theta_A}{\ell}.$$

Q. **A straight rod AB having length ℓ is kept inclined to a wall as shown in figure. End A of the rod is moving towards right with a speed v_A. Calculate-**
(A) Speed of end B of the rod at the given instant
(B) Angular velocity of rod at the given instant

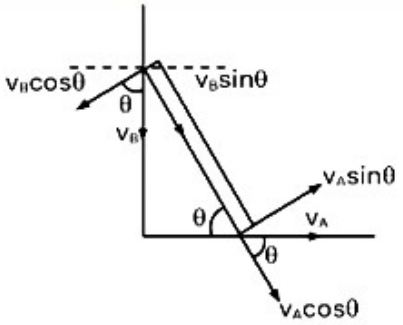


Sol. (A) Consider the speed of end B of the rod is v_B in downward direction.

According to the rigid body constraint, the component of velocity of A and B along AB must be equal.

$v_B\sin\theta = v_A\cos\theta$

$$\Rightarrow \quad v_B = v_A\frac{\cos\theta}{\sin\theta} = v_A\cot\theta$$



(B) The velocity components perpendicular to the rod produce rotation.

$$\text{So, } \omega = \frac{v_B \cos\theta + v_A \sin\theta}{\ell}$$

$$= \frac{\left(v_A \frac{\cos\theta}{\sin\theta}\right)\cos\theta + v_A \sin\theta}{\ell}$$

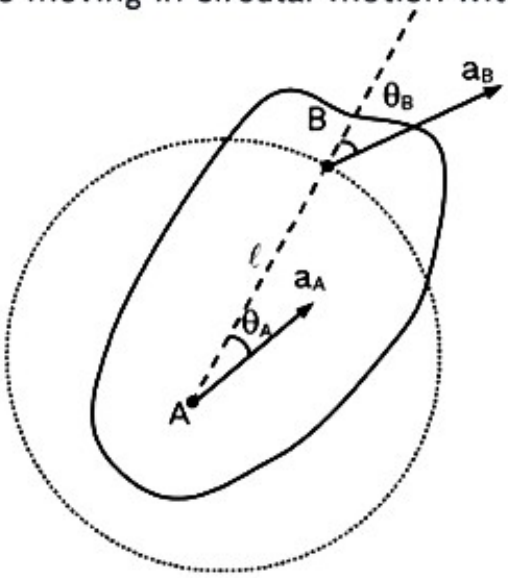
$$\omega = \frac{v_A\left[\cos^2\theta + \sin^2\theta\right]}{\ell \sin\theta} = \frac{v_A}{\ell \sin\theta}$$

Rigid Body Constraint for Acceleration:

Similar to the rigid body constraint of velocity, we can also find relations for acceleration and angular acceleration of different points on a rigid body.

Consider the situation shown below where two points A and B on a rigid body are having accelerations a_A and a_B respectively.

Now, B can be assumed to be moving in circular motion with respect to A,



Then, relative acceleration along the line joining the two points should be equal to the centripetal acceleration.

i.e., $a_A \cos\theta_A - a_B \cos\theta_B = \omega^2 \ell$ ..... (Normal or centripetal acceleration)

Where, ω is the angular velocity of the body and ℓ is the distance between A and B.

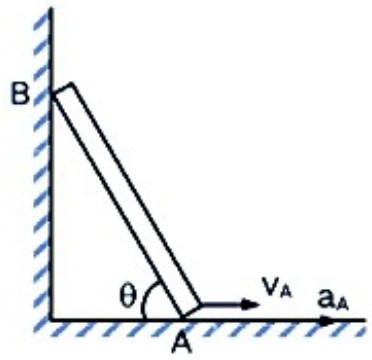
And, the relative acceleration perpendicular to the line joining A and B gives the tangential acceleration.

$$\left|a_A \sin\theta_A - a_B \sin\theta_B\right| = \ell\alpha \quad \text{.....(tangential acceleration of B wrt A)}$$

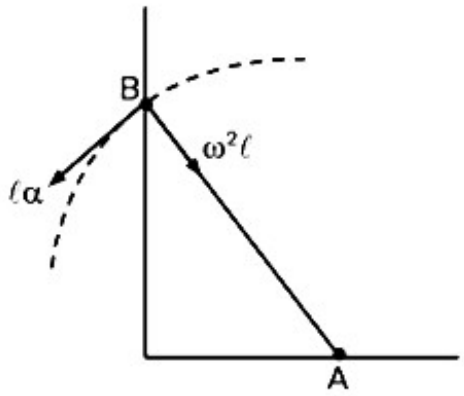
Where, α is the angular acceleration of the body.

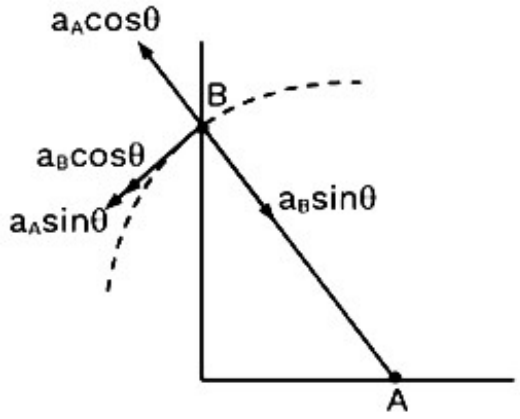
Similar to angular velocity, angular acceleration is also same for all the points on the rigid body.

Q. **A straight rod AB of length ℓ is kept inclined to a vertical wall as shown in figure. At the given time instant, end A of the rod is having a speed v_A towards right and acceleration a_A towards right as shown. Calculate the acceleration of end B of the rod and angular acceleration of the rod.**



Sol. From the frame of reference of end A of the rod, we can draw the below diagram.





Acceleration of end B wrt ground, $\vec{a}_{B/G} = \vec{a}_{B/A} + \vec{a}_{A/G}$

Also, $\omega^2\ell = a_B \sin\theta - a_A \cos\theta$

and, $\ell\alpha = a_A \sin\theta + a_B \cos\theta$

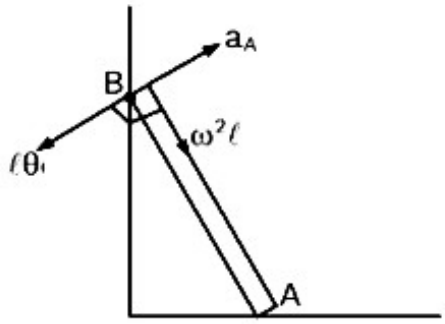
Now, $\omega = \dfrac{v_A}{\ell \sin\theta}$

Thus, $a_B = \left[\dfrac{v_A^2}{\ell \sin^2\theta} + a_A \cos\theta\right]\dfrac{1}{\sin\theta}$

Angular acceleration, $\alpha = \frac{1}{\ell}\left[a_A \sin\theta + \left[\frac{v_A^2}{\ell \sin^2\theta} + a_A \cos\theta\right]\cot\theta\right]$

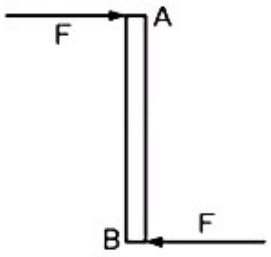
Also, $\omega^2\ell = \frac{v_A^2}{\ell \sin^2\theta}$

Now, from reference frame of ground, net acceleration of point B is the vector sum of all the three accelerations shown in figure.



Torque

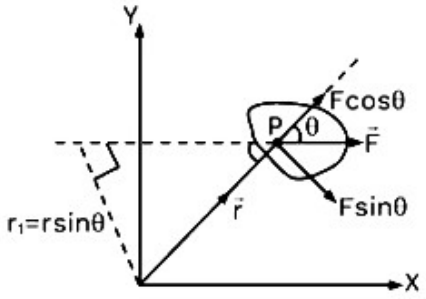
Rotational effect of a force is termed as torque.
If a rod AB is acted upon by two equal and opposite forces as shown in the figure below, then the rod will turn (rotate) but net force on the rod is zero.



From above example, we can say that bodies can rotate without any net external force. It can also be said that for a body to rotate, something other than force is required. That is called "Torque".
Torque is the rotational analogue of force or turning effect of force. It is the measure of the tendency of a force to rotate an object about some axis. Torque is represented by τ. Its SI unit is N-m and dimensional formula is $[M\ L^2\ T^{-2}]$. Note that the unit of "Torque" is same as unit of "Work" but these two are entirely different quantities and are not to be confused as same in study of mechanics.

Torque of a Force about a Point:

Let's take an arbitrary body as shown in the figure. Let's say that a force $\vec{F}$ is applied at a point P on the body whose position vector is $\vec{r}$ as shown in the figure.



The torque of this force (F) about origin is defined as,

$$\vec{\tau} = \vec{r} \times \vec{F}$$

Direction of torque can be found by using the rule of cross-product. Keep palm of right hand along $\vec{r}$ and rotate it towards $\vec{F}$. Then, the thumb gives the direction of $\vec{\tau}$.
Let the angle between $\vec{r}$ and $\vec{F}$ be θ,

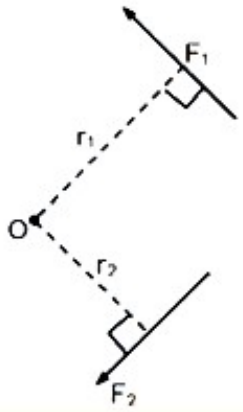
Then, $|\vec{\tau}| = rF\sin\theta$

$|\vec{\tau}| = rF_{\perp}$ ($F_{\perp}$ is the component of $\vec{F}$ perpendicular to $\vec{r}$)

$|\vec{\tau}| = Fr_{\perp}$ ($r_{\perp}$ is the component of $\vec{r}$ perpendicular to $\vec{F}$)

Where, $r_{\perp} = r\sin\theta$ is called the moment arm (or the lever arm) of $\vec{F}$.

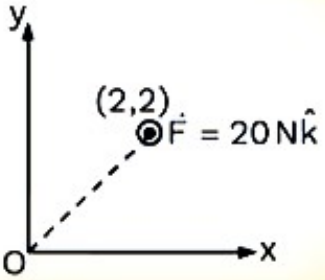
Q. **Find the torque due to forces F_1 and F_2 about point O in the figure given below.**



Sol. About point O, F_1 gives anticlockwise torque while F_2 gives clockwise torque. Taking anticlockwise direction as +ve and clockwise direction as –ve,

$$\tau_0 = F_1r_1 - F_2r_2$$

Q. **A force of magnitude 20 N is acting at point (2m, 2m) in +z-direction as shown.**
What is
(a) torque due to the given force about point O.
(b) torque due to the given force about x-axis.



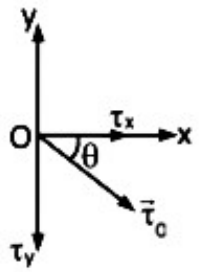
Sol. (a) $\vec{\tau}_0 = \vec{r} \times \vec{F}$

$= [2\hat{i} + 2\hat{j}] \times [20\hat{k}]$

$= 40(\hat{i} \times \hat{k}) + 40(\hat{j} \times \hat{k})$

$= 40(-\hat{j}) + 40\hat{i}$

$|\vec{\tau}_0| = \sqrt{(-40)^2 + 40^2} = 40\sqrt{2}$ Nm



(b) $\tau_x = 40$ Nm

Vector form of torque:

If a 3-dimensional force is acting at a point having a 3-dimensional position vector, then

For $\vec{F} = F_x\hat{i} + F_y\hat{j} + F_z\hat{k}$

$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$

$$\vec{\tau} = \vec{r} \times \vec{F} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

Cross product is calculated using determinant method.

Using determinant method,

$$\vec{\tau} = [\tau_x\hat{i} + \tau_y\hat{j} + \tau_z\hat{k}] = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ F_x & F_y & F_z \end{vmatrix}$$

$$= \hat{i}[yF_z - zF_y] - \hat{j}[xF_z - zF_x] + \hat{k}[xF_y - yF_x]$$

$$= \hat{i}[yF_z - zF_y] + \hat{j}[zF_x - xF_z] + \hat{k}[xF_y - yF_x]$$

so, $\tau_x = yF_z - zF_y$

$\tau_y = zF_x - xF_z$

$\tau_z = xF_y - yF_x$

If object is moving in y-z plane, $\tau_x \neq 0$, $\tau_y = 0$, $\tau_z = 0$

If object is moving in x-z plane, $\tau_y \neq 0$, $\tau_x = \tau_z = 0$

If object is moving in x-y plane, $\tau_z \neq 0$, $\tau_x = \tau_y = 0$

Q. **A 3-dimensional force $(5\hat{i} - 4\hat{j} + 6\hat{k})$ N acts on a particle having a position vector $(\hat{i} + 2\hat{j} + 3\hat{k})$, calculate the torque acting on the particle about the origin.**

Sol. Torque, $\vec{\tau} = \vec{r} \times \vec{F}$

$$\vec{\tau} = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ 1 & 2 & 3 \\ 5 & -4 & 6 \end{vmatrix}$$

$$= \hat{i}[12 - (-12)] - \hat{j}[6 - 15] + \hat{k}[-4 - 10] = (24\hat{i} + 9\hat{j} - 14\hat{k})\text{Nm}$$

Equilibrium of a Rigid Body

If the linear momentum and angular momentum of a rigid body do not change with respect to time, then the rigid body is said to be in equilibrium.

In such a case, linear acceleration and angular acceleration of the body are zero.

Then, for equilibrium of a rigid body,

(a) The vector sum of all the forces acting on the rigid body is zero,

$$\vec{F}_1 + \vec{F}_2 + \vec{F}_3 + ... + \vec{F}_n = \sum_{i=1}^{n} \vec{F}_i = 0$$

For total force on the rigid body to be zero, the total linear momentum of the body remains constant with respect to time.

Above equation shows the condition of translational equilibrium of rigid body.

(b) For vector sum of all the torques acting on the rigid body to be zero,

$$\vec{\tau}_1 + \vec{\tau}_2 + \vec{\tau}_3 + ... + \vec{\tau}_n = \sum_{i=1}^{n} \vec{\tau} = 0$$

The total angular momentum of the body remains constant with respect to time. Above equation shows the condition for the rotational equilibrium of the rigid body.

Q. **A straight and uniform wooden stick having mass of 1.6 kg and length l is resting in an inclined manner on a vertical and smooth wall whose height is h (< l). A small portion of the stick is extended beyond the wall. The reaction force of the wall on the stick is directed perpendicular to the stick.**
The stick is making an angle of 30° with the wall and the lower end of the stick is kept on a rough floor. The reaction of the wall on the stick is equal in magnitude to the reaction of the floor on the stick. The ratio h/l and the frictional force f at the bottom of the stick are ($g = 10\ ms^{-2}$)

(A) $\frac{h}{l} = \frac{\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} N$

(B) $\frac{h}{l} = \frac{3}{16}, f = \frac{16\sqrt{3}}{3} N$

(C) $\frac{h}{l} = \frac{3\sqrt{3}}{16}, f = \frac{8\sqrt{3}}{3} N$

(D) $\frac{h}{l} = \frac{3\sqrt{3}}{16}, f = \frac{16\sqrt{3}}{3} N$

Sol. Ans. (D)

Given situtation is shown in the adjacent figure.

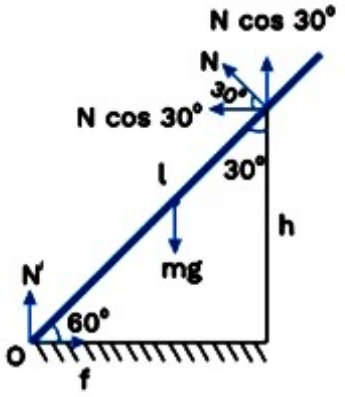
Here m = 1.6 kg, g = 10 ms^{-2}

Also, N' = N ...(given)

For translation equilibrium,

In Horizontal direction, f = N cos 30°

In Vertical direction, N' + N sin 30° = mg



$$N + \frac{N}{2} = 1.6 \times 10$$

$$\frac{3N}{2} = 16 \quad \Rightarrow N = \frac{32}{3}N$$

Now,

$$f = \frac{32}{3} \times \cos 30° = \frac{16\sqrt{3}}{3}N$$

Stick is also in rotational equilibrium, so net torque about point O will be zero.

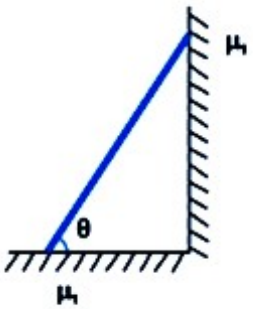
$$\tau_O = 0$$

$$\Rightarrow -(mg)\left(\frac{l}{2}\cos 60°\right) + N\left(\frac{h}{\sin 60°}\right) = 0$$

$$\Rightarrow mg\left(\frac{l}{4}\right) = \frac{2Nh}{\sqrt{3}} \quad \Rightarrow \frac{h}{l} = \frac{\sqrt{3}mg}{8N}$$

$$\Rightarrow \frac{h}{l} = \frac{\sqrt{3} \times 1.6 \times 10}{8 \times \frac{32}{3}} = \frac{3\sqrt{3}}{16}$$

Q. **In the figure, a uniform rod of mass m is shown leaning against a wall. The rod is in static equilibrium condition while it makes an angle θ with the horizontal floor. The value of friction coefficient between the wall and the rod is μ_1 and that between the floor and the rod is μ_2. The normal reaction on the rod due to wall is N_1 and that due to the floor is N_2. If the rod is at limiting equilibrium, then**



(A) $\mu_1 = 0, \mu_2 \neq 0$, and $N_2 \tan\theta = \frac{mg}{2}$

(B) $\mu_1 \neq 0, \mu_2 = 0$, and $N_1 \tan\theta = \frac{mg}{2}$

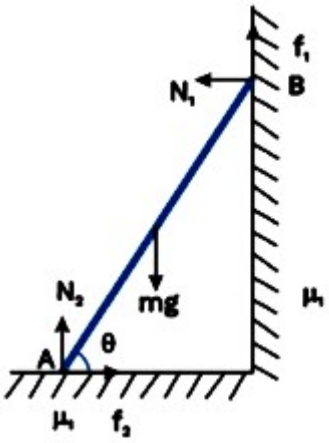
(C) $\mu_1 \neq 0, \mu_2 \neq 0$, and $N_2 = \frac{mg}{1+\mu_1\mu_2}$

(D) $\mu_1 = 0, \mu_2 \neq 0$, and $N_1 \tan\theta = \frac{mg}{2}$

Sol. Ans (C, D)

The various forces acting on the rod are as shown in the figure. Let L be the length of the rod. As the rod is about to slip, so both frictions will be limiting.

$\therefore f_1 = \mu_1 N_1$



And, $f_2 = \mu_2 N_2$

In options (a) and (d), μ =

$\therefore f_1 = 0$

For rotational equilibrium,

Taking the moments about end A of the rod,

$$N_1 L \sin\theta - mg\frac{L}{2}\cos\theta = 0$$

$$N_1 \sin\theta = \frac{mg}{2}\cos\theta \text{ or } N_1 \tan\theta = \frac{mg}{2}$$

Hence, option (d) is correct.

For translational equilibrium,

Taking the forces in horizontal and vertical direction,

$f_2 - N_1 = 0 \quad \Rightarrow N_1 = f_2 = \mu_2 N_2$, and $\quad N_2 - mg = 0 \quad \Rightarrow N_2 = mg$

Hence, option (a) is not correct.

In option (b), $\mu_2 = 0$

There is no force to balance N_1, so rod cannot remain in equilibrium.

Hence, option (b) is not correct.

In option (c), $\mu_1 \neq 0, \mu_2 \neq 0$

For translational equilibrium,

Taking the forces in horizontal and vertical directions,

$$f_2 - N_1 = 0 \qquad \text{..... (i)}$$

And, $N_2 + f_1 - mg = 0$ ..... (ii)

From eqn. (i),

$$N_1 = f_2 = \mu_2 N_2 \qquad \text{..... (iii)}$$

From eqn. (ii), $N_2 + f_1 = mg$

$$N_2 + \mu_1 N_1 = mg \qquad \text{..... (iv)}$$

$$N_2 + \mu_1(\mu_2 N_2) = mg \qquad \text{(Using (iii))}$$

$$N_2(1 + \mu_1\mu_2) = mg$$

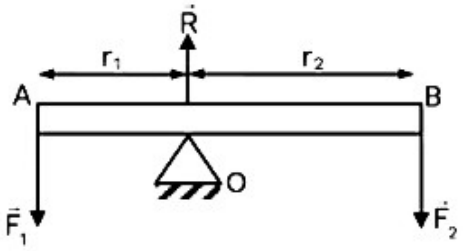
$$N_2 = \frac{mg}{(1+\mu_1\mu_2)}$$

Hence, option (c) is correct

Principle of Moments

A straight rod having negligible mass which is pivoted at any point along its length is called a lever. The point about which the rod is pivoted is called fulcrum.

In the figure shown below,



Let $\vec{R}$ be the support reaction at the fulcrum.

For translational equilibrium, $R - F_1 - F_2 = 0$

For rotational equilibrium, $\sum \tau_0 = 0$

So, $F_1 r_1 - F_2 r_2 = 0$

$F_1 r_1 = F_2 r_2$

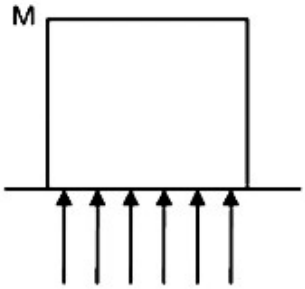
The above equation gives the principle of moments for a lever.

The ratio of $\frac{F_1}{F_2}$ is called "Mechanical Advantage (M.A.)"

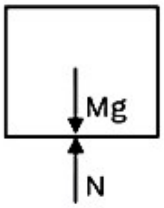
$$\text{M.A.} = \frac{F_1}{F_2} = \frac{r_2}{r_1}$$

Shift in Normal Reaction and Toppling

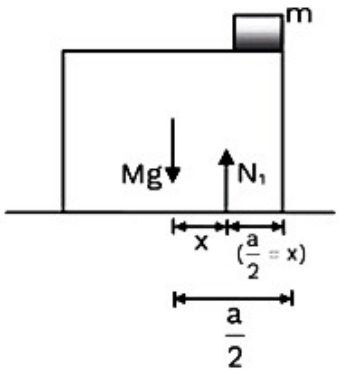
For a block kept on horizontal surface, there will be normal reaction on it due to ground in upward direction. For uniform mass distribution of the block, the normal reaction will be uniformly distributed on the whole surface area of the block.



To balance the torque and weight of the block, normal reaction should have magnitude equal to weight of block and normal reaction will be passing effectively through centre of block.



On placing a small mass m at one edge of the block, normal reaction will shift from centre of block to some position towards right, so that the net torque can be balanced on the block.



For a shift of x in the normal reaction, for translational equilibrium,

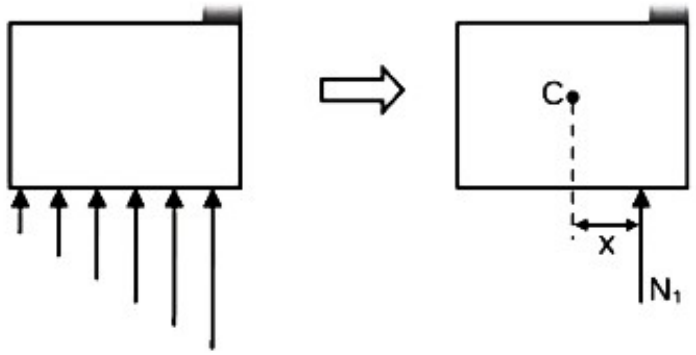
$N_1 = (M + m)g$

And, for rotational equilibrium,

$$Mgx - mg\left(\frac{a}{2} - x\right) = 0$$

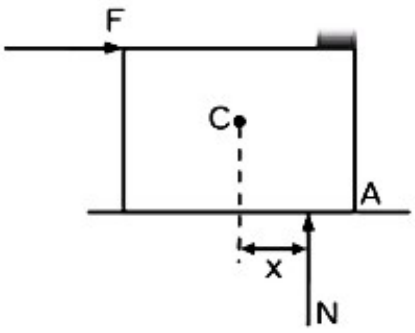
$$x = \frac{mg.\frac{a}{2}}{(M+m)g} = \frac{ma}{2(M+m)}$$

Note that, in practical condition, normal reaction cannot act at a single point, but we consider it to be passing through a single point effectively.
After the shift in the normal reaction, the normal reaction is still distributed but its distribution is non-uniform.



Toppling

A force is being applied tangentially on a block as shown in figure. The torque due to force F and frictional force f about centre of mass should be zero for equilibrium.
Now, on increasing the force F, the normal reaction will shift further towards right. But the normal reaction cannot shift beyond point A.
Now, if we increase the force F further, then normal reaction cannot shift further to balance the torque.
Then the block will start rotating about point A. This is called toppling of the block.



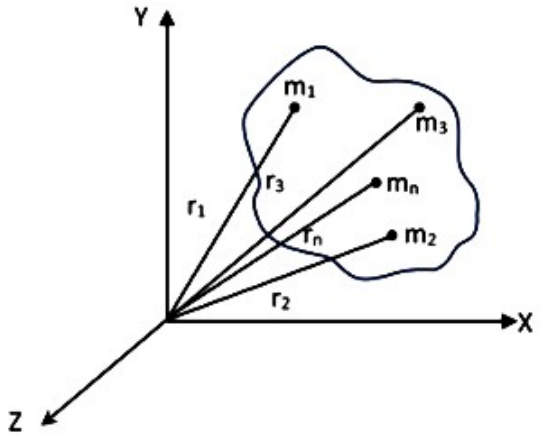
Moment of Inertia

In linear motion, the mass of a body is the measure of its inertia (resistance to linear motion). Similarly, in rotational motion, the moment of inertia about a given axis of rotation is the measure of resistance to a change in its rotational motion.
Moment of inertia is a measure of the way in which different masses in a body are distributed at different distances from the axis of rotation.

Moment of inertia depends upon:

a. The mass of the body,
b. The shape and size of the body,
c. The distribution of mass about the axis of rotation, and
d. The position and orientation of the axis of rotation.

For a rigid body consisting of n particles having masses m_1 , m_2 , m_3 , ...m_n at distances r_1, r_2, r_3,.....r_n from the axis of rotation, the moment of inertia is given by,



$$I = m_1r_1^2 + m_2r_2^2 + m_3r_3^2 + ... + m_nr_n^2 = \sum_{i=1}^{n} m_ir_i^2$$

The dimensional formula of moment of inertia is [M L^2 T^0] and its S.I. unit is kg m^2.

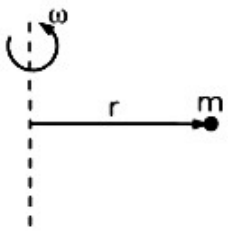
Calculation of Moment of Inertia

For a point mass

Moment of inertia, $I = mr^2$

m = Mass of point mass

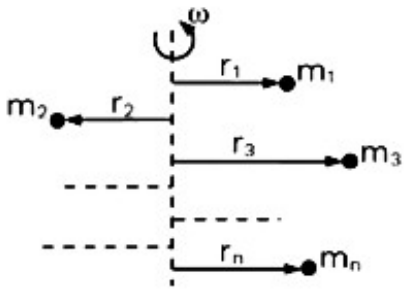
r = Distance of the point mass from the axis of rotation



For a system of n-particles

$$I = m_1r_1^2 + m_2r_2^2 + ... + m_nr_n^2$$

$$I = \sum_{i=1}^{n} m_ir_i^2$$



Note: Moment of Inertia is a scalar quantity and for a number of particles, it follows scalar addition, if all the moments of inertia are calculated about the same axis.

For systems with continuous mass arrangement

If in a system, mass is distributed uniformly, then we have to follow the integration approach to find the total moment of inertia of the body.

We take an inifinitesimal mass element dm at perpendicular distance r from the axis or rotation.

Then, we write the moment of inertia of the small mass dm as,

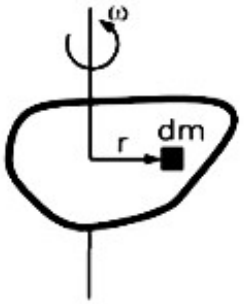
$dI = r^2\, dm$

Now, we can integrate the above expression to find the total moment of inertia of the body.

$dI = r^2 dm$

Now, we can integrate the above expression to find the total moment of inertia of the body.

$\Rightarrow I = \int dI = \int r^2 dm$, (where r is the perpendicular distance from axis)



Moment of Inertia of some Uniform Rigid Bodies

1. **Uniform Ring (Mass = M, Radius = R)**

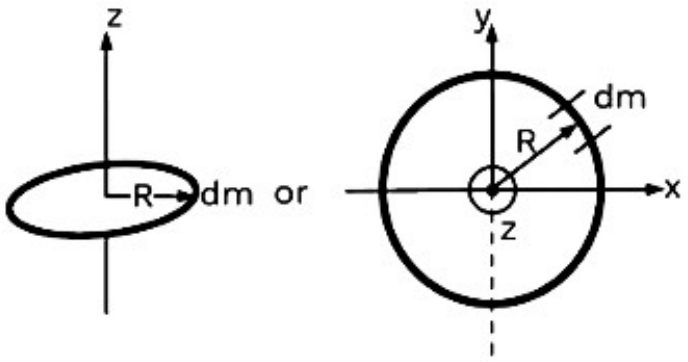
   We can assume a small elemental mass on the periphery of the ring.

   Moment of inertia of this small elemental mass dm about an axis passing through the centre of the ring and perpendicular to its plane will be, $dI = \int dmR^2$

   Now, we can integrate this expression to find moment of inertia of the complete ring about the given axis,

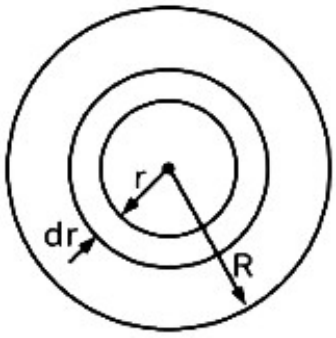
   $I_z = \int R^2 dm = R^2 \int_0^M dm$

   $I_z = MR^2$



Moment of inertia of a uniform hollow cylinder of mass M and radius R is same as that of a uniform ring of mass M and radius R about an axis passing through its centre and perpendicular to its plane.

2. **Uniform disc (Mass = M, Radius = R)**
To find the moment of inertia of a solid uniform disc about an axis passing through its centre and perpendicular to its plane, we can consider an elemental ring having radius 'r' and thickness dr as shown in the figure.



Then, moment of inertia of the elemental ring about the axis, $dI = \int dmr^2$

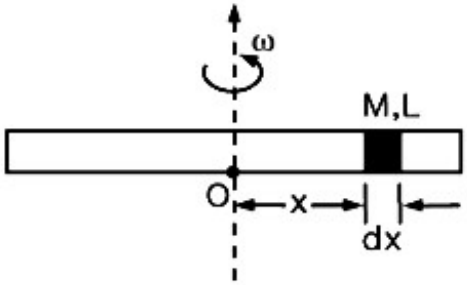
Now, we can integrate above expression to find the moment of inertia of the complete disc, $I = \int_0^R dmr^2$

$$= \int_0^R \left(\frac{M}{\pi R^2} 2\pi r dr\right) r^2 = \frac{2M}{R^2} \int_0^R r^3 dr = \frac{MR^2}{2}$$

Moment of inertia of a uniform solid cylinder of mass M and radius R is same as that of a uniform disc of mass M and radius R about an axis passing through its centre and perpendicular to its plane.

3. **Uniform Rigid Rod (Mass = M, Length = L)**

(a) Moment of inertia about an axis passing through centre of mass of the rod & perpendicular to the length of rod.



$$I = \int x^2 dm$$

$$= \int_{-L/2}^{+L/2} x^2 \frac{M}{L} dx \qquad \left[\because dm = \frac{M}{L} dx\right]$$

$$= \frac{M}{L}\int_{-L/2}^{+L/2} x^2 dx = \frac{M}{3L}[x^3]_{-L/2}^{+L/2}$$

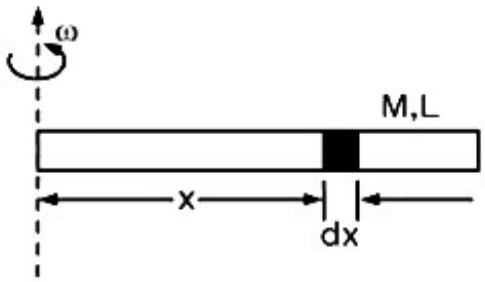
$$\Rightarrow \qquad I = \frac{1}{12} ML^2$$

(b) Moment of inertia about an axis passing through one end of the rod & perpendicular to its length.

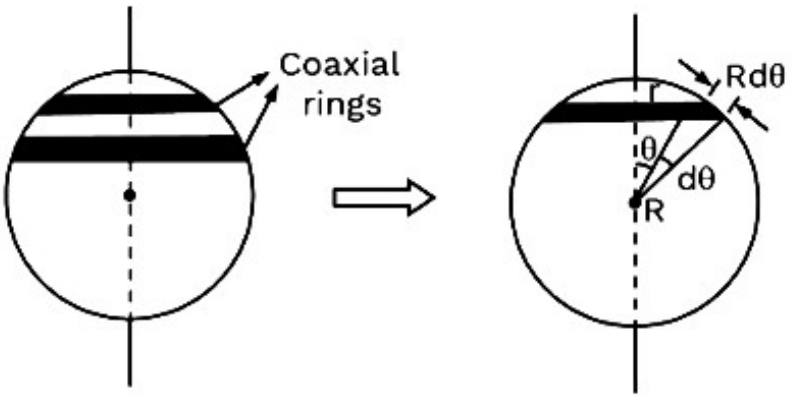
$$I = \int_0^L x^2 dm$$

$$= \int_0^L x^2 \frac{M}{L} dx$$

$$I = \frac{1}{3} ML^2$$



4. **Hollow sphere (M, R)**



To find the moment of inertia of a hollow sphere about one of its diameters, we divide it in an infinite number of co-axial rings. Each ring is infinitesimal having mass dm and the radius of all the rings is different.
If the moment of inertia of one general ring is dI, then moment of inertia of the hollow sphere will be the sum of moment of inertia of all coaxial rings. We can simply use integration approach to find the total moment of inertia of the hollow sphere.

$$\Rightarrow \quad I = \int dI = \int r^2 dm$$

dm = mass of ring

= Mass per unit area × Area

$$= \left(\frac{M}{4\pi R^2}\right) \times 2\pi r R d\theta = \frac{M}{2R^2}.rRd\theta$$

$$= \frac{M}{2R}.rd\theta = \frac{M}{2}\sin\theta d\theta \qquad \left(\because \sin\theta = \frac{r}{R}\right)$$

$$I = \int R^2 \sin^2\theta \frac{M}{2}\sin\theta d\theta$$

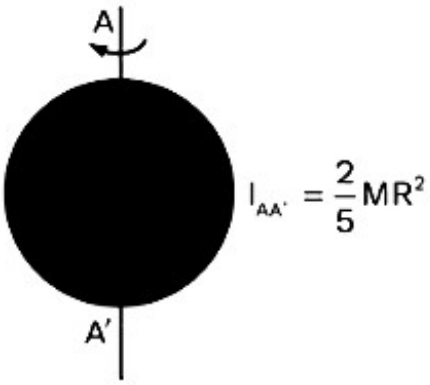
When θ is varied by 0 to 90°, then half of the hollow sphere is covered. So, we can vary θ from −90° to +90°, or from 0 to 180° to get moment of inertia of the complete hollow sphere. So,

$$= \frac{M}{2}R^2\int_0^{\pi} \sin^3\theta d\theta = \frac{M}{2}R^2\int_0^{\pi}(1-\cos^2\theta)\sin\theta d\theta$$

$$= \frac{MR^2}{2}\int_{1}^{-1}(1-t^2)(-dt) \qquad (\text{Let } \cos\theta = t,\ -\sin\theta\, d\theta = dt)$$

$$= \frac{-MR^2}{2}\int_{1}^{-1}(1-t^2)dt = \frac{2MR^2}{3}$$

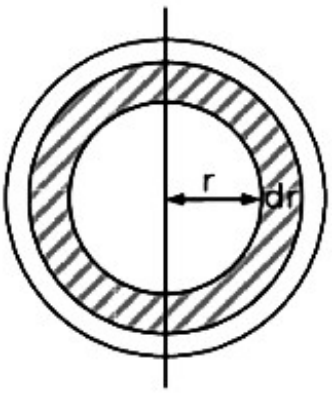
5. **Solid Sphere (Mass = M, Radius = R)**



A solid sphere can be divided into infinite number of concentric hollow spheres. Consider a hollow sphere of radius r and thickness dr. Let moment of inertia of hollow sphere is dI, then moment of inertia of solid sphere is the sum of moments of inertia of all the hollow spheres.

Moment of inertia of an elemental hollow sphere is, $dI = \frac{2}{3}(\text{mass})\, r^2$

But, mass = density × volume = $\rho \times 4\pi r^2 dr$



$$\Rightarrow dI = \frac{8}{3}\pi\rho r^4\, dr$$

$$\Rightarrow I = \int_0^R dI = \frac{8\pi\rho}{3}\int_0^R r^4 dr = \frac{8\pi\rho}{3}\times\frac{R^5}{5}$$

$$\Rightarrow I = \frac{8\pi}{3}\times\frac{M}{\frac{4\pi R^3}{3}}\times\frac{R^5}{5} = \frac{2}{5}MR^2 \qquad \left(\because \rho = \frac{\text{Mass}}{\text{Volume}} = \frac{M}{\frac{4}{3}\pi R^3}\right)$$

Q. **Two metallic discs having densities ρ_1 and ρ_2 have same mass and same thickness. What is the ratio of their moment of inertia?**

Sol. For the same mass and same thickness of the both discs,

$$M_1 = M_2$$

$$\rho_1 \left(\pi R_1^2 t\right) = \rho_2 \left(\pi R_2^2 t\right)$$

$$\frac{R_1^2}{R_2^2} = \frac{\rho_2}{\rho_1}$$

The ratio of moment of inertia will be, $\frac{I_1}{I_2} = \frac{\left(\frac{MR_1^2}{2}\right)}{\left(\frac{MR_2^2}{2}\right)} = \frac{R_1^2}{R_2^2} = \frac{\rho_2}{\rho_1}$

Q. **An arc making an sector of 90° is removed from a circular ring of mass m and radius R. If the moment of inertia of the remaining part of the ring about an axis passing through the centre of the ring and perpendicular to its plane is kmR^2, then find the value of k.**

Sol. If ring corresponding to a sector of 90° is removed, then one-fourth of the mass of the ring is removed.

Remaining mass, $m' = \frac{3}{4}m$

All the remaining mass is still at the same radial distance from the axis.

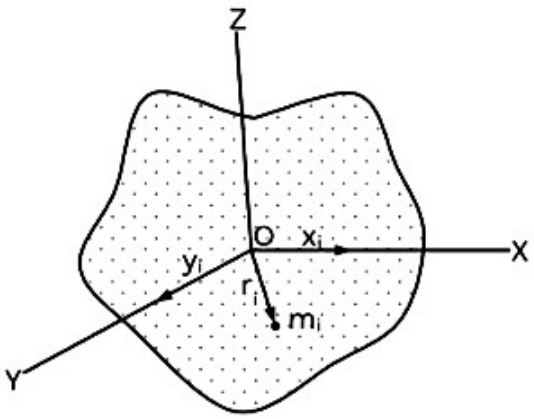
So, moment of inertia of the remaining ring = $I = m'R^2 = \frac{3}{4}mR^2$

So, k = 3/4.

Perpendicular and Parallel Axis Theorems

Perpendicular Axes Theorem

As per perpendicular axis theorem, the moment of inertia of a planar body (2D body) about an axis perpendicular to its plane is equal to the sum of its moments of inertia about two perpendicular axes concurrent with the perpendicular axis and lying in the plane of the body.



Let I_x, I_y and I_z be the moment of inertia the arbitrary shaped plate about the axes OX, OY and OZ respectively.

Now $I_x = m_1y_1^2 + m_1y_2^2 + ...m_ny_n^2$

$$\Rightarrow I_x = \sum_{i=1}^{n} m_i y_i^2 \qquad ...(i)$$

Similarly, $I_y = \sum_{i=1}^{n} m_i x_i^2$ ...(ii)

and $I_z = \sum_{i=1}^{n} m_i r_i^2$ ...(iii)

Adding equations (i) and (ii), we get

$$I_x + I_y = \sum_{i=1}^{n} m_i (y_i^2 + x_i^2)$$

In the above figure, by Pythagoras theorem

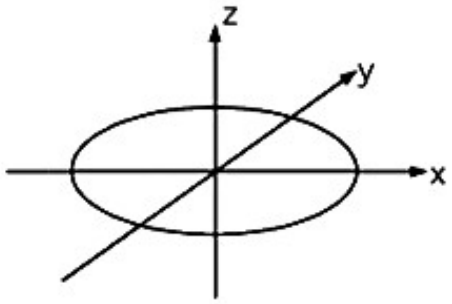
$$r_i^2 = x_i^2 + y_i^2$$

So, $$I_x + I_y = \sum_{i=1}^{n} m_i r_i^2 = I_z$$

Thus, $I_z = I_x + I_y$

Q. Find the moment of inertia of a uniform disc about its diameter?

Sol.



Moment of inertia of a uniform disc about an axis passing through its centre and perpendicular to its plane is given by,

$$I_z = (MR^2)/2$$

As per perpendicular axis theorem,

$$I_z = I_x + I_y$$

From the figure, x and y axes drawn along the diameter of the disc, then by symmetry,

$$I_x = I_y$$

$$\therefore \quad I_z = 2I_x$$

$$\therefore \quad I_x = \frac{1}{2}\left(\frac{MR^2}{2}\right) = \frac{MR^2}{4}$$

So, moment of inertia of the disc about any of its diameter is $\frac{MR^2}{4}$.

Parallel axis theorem

As per parallel axis theorem, the moment of inertia of a body about any axis is equal to the sum of the moment of inertia of the body about a parallel axis passing through its centre of mass and the product of its mass and the square of the distance between the two parallel axes.

The theorem of parallel axis is applicable to a body of any shape.



Consider that AB axis is passing through the center of mass of the arbitrary shaped body as shown. Another axis EF is at a parallel distance d from the axis AB. Then, as per parallel axis theorem,

$$I_{EF} = I_{AB} + Md^2$$

Q. What is the moment of inertia of a rod of mass M and length L about an axis perpendicular to it and passing through its one end?

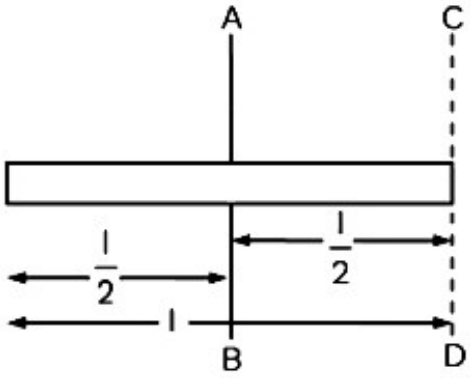
Sol. For the rod of mass M and length L, the moment of inertia of the rod about an axis AB passing through its centre of mass is given by, $I_{AB} = \frac{ML^2}{12}$

As per parallel axis theorem,

$$I_{CD} = I_{AB} + M\left(\frac{L}{2}\right)^2$$

$$= \frac{ML^2}{12} + \frac{ML^2}{4} = \frac{ML^2 + 3ML^2}{12} = \frac{4ML^2}{12} = \frac{ML^2}{3}$$

$$I_{CD} = \frac{ML^2}{3}$$



Q. The moment of inertia of a uniform cylinder having length L and radius R about its perpendicular bisector is I. What is the ratio L/R so that the moment of inertia is minimum?

Sol. Moment of inertia of a uniform cylinder about its perpendicular bisector will be,

$$I = \frac{MR^2}{4} + \frac{ML^2}{12}$$

Lets convert R in terms of L now.

Mass of the cylinder, $M = \rho\left(\pi R^2 L\right)$

So, $R^2 = \frac{M}{\pi\rho L}$

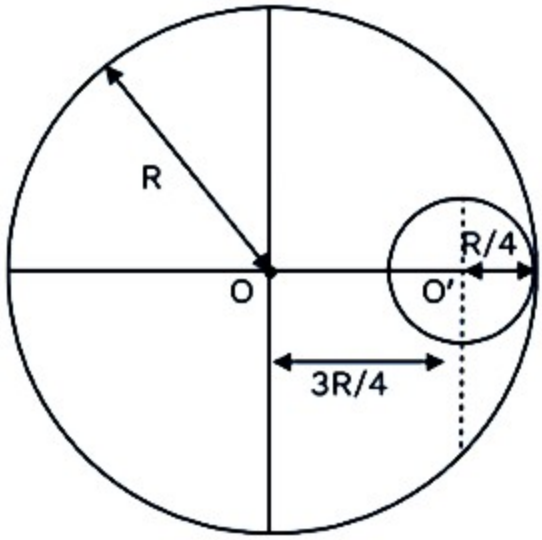
Then, we can write moment of inertia as, $I = \frac{M}{4}\left(\frac{M}{\pi\rho L}\right) + \frac{ML^2}{12} = \frac{M}{4}\left(\frac{M}{\pi\rho L} + \frac{L^2}{3}\right)$

For I to be minimum, $\frac{dI}{dL} = 0$

$$-\frac{M}{\pi\rho L^2} + \frac{2L}{3} = 0 \text{, then, } \frac{2L}{3} = \frac{M}{\pi\rho L^2} = \frac{R^2}{L}$$

$$\frac{L^2}{R^2} = \frac{3}{2} \Rightarrow \boxed{\frac{L}{R} = \sqrt{\frac{3}{2}}}$$

Q. **From a thin uniform disc of mass M and radius R, a circular hole of radius R/4 is cut as shown in the figure.**



What is the moment of inertia of the remaining portion of the disc about an axis passing through the point O and perpendicular to the plane of the disc?

Sol. Total moment of inertia of the complete disc about an axis passing through O and perpendicular to the plane of the disc, $I_o = \frac{MR^2}{2}$

Mass of the removed portion from the disc, $M' = \frac{M}{\pi R^2} \times \pi\left(\frac{R}{4}\right)^2 = \frac{M}{16}$

Moment of inertia of the small removed disc about the axis passing through O,

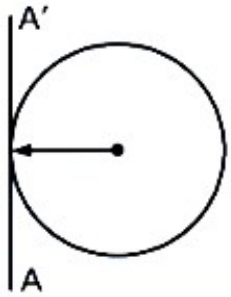
$$I' = \frac{\left(\frac{M}{16}\right)\left(\frac{R}{4}\right)^2}{2} + \left(\frac{M}{16}\right)\left(\frac{3R}{4}\right)^2 = \frac{MR^2}{512} + \frac{9MR^2}{256} = \frac{19}{512}MR^2$$

So, moment of inertia of the remaining portion about the axis passing through O and perpendicular to its plane, $I_f = I_0 - I' = \frac{MR^2}{2} - \frac{19}{512}MR^2 = \frac{237}{512}MR^2$

Radius of Gyration : Radius of gyration is the radial distance from the given axis of rotation where the entire mass of the body can be assumed to be concentrated so that its moment of inertia does not change.

$K = \sqrt{\frac{I}{m}}$, where I is the moment of inertia.

Q. **Calculate the radius of gyration of a uniform disc of mass 'm' and radius R about an axis passing tangentially to the disc.**



Sol. Moment of inertia (I) of uniform solide disc about AA'

$$= \frac{mR^2}{4} + mR^2$$

$$= \frac{5mR^2}{4}$$

Radius of gyration, $K = \sqrt{\frac{I}{m}} = \sqrt{\frac{5mR^2}{4m}} = \sqrt{\frac{5R^2}{4}} = \sqrt{\frac{5}{4}}R$

Q. **A disc of mass m and radius R is melted and it is reshaped into a ring of radius 2R keeping the mass same. By what factor the radius of gyration changes in the process about an axis passing through the centre and perpendicular to the surface?**

Sol. For a disc of mass m and radius R,

$$\frac{mR^2}{2} = mk_1^2 \quad \Rightarrow k_1 = \frac{R}{\sqrt{2}}$$

For the ring of mass m and radius 2R,

$$m(2R)^2 = mk_2^2 \quad \Rightarrow k_2 = 2R$$

Now, $\frac{k_2}{k_1} = \frac{2R}{\left(\frac{R}{\sqrt{2}}\right)} = 2\sqrt{2}$

So, radius of gyration increases by a factor of $2\sqrt{2}$.

Dynamics of Rotational Motion About a Fixed Axis

Laws of motion for rotation can be derived in similarity with the laws of motion in translation.

In linear motion, force, $F = ma$

In rotational motion, torque, $\tau = I\alpha$

In linear motion, work, $W = F.dx$, and

In rotational motion about a fixed axis, work, $W = \tau.d\theta$

Power is the rate of work done i.e., $P = \frac{dW}{dt}$

Hence, we get $P = \frac{\tau.d\theta}{dt} = \tau\omega$

Above relation gives instantaneous power.

Kinetic energy of rotational motion is given by, $K = \frac{1}{2}I\omega^2$

Q. **Find the value of instantaneous power of a wheel which is rotating with angular velocity of 15 rad/s, when a torque of 10 Nm is applied to it.**

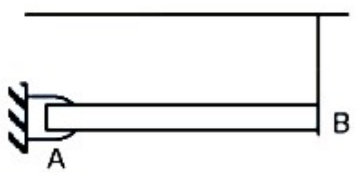
Sol. Instantaneous power, $P = \tau\omega$

Torque, $\tau = 10$ Nm

$\omega = 15$ rad/s

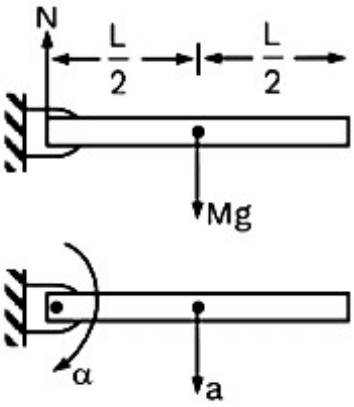
$\therefore$ $P = 10 \times 10 = 150$ W

Q. **A uniform rod AB of length L and mass M is held horizontally with the help of one light string and a hinge as shown. If string at B is cut suddenly, find out the initial angular acceleration of the rod and the initial reaction force at the hinge.**



Sol. Just after cutting the string at B, Mg produces a torque about A.

$$\tau_A = Mg\frac{L}{2}$$



Also, $\tau = I\alpha$

So, $I_A \alpha = Mg\frac{L}{2}$

For rod, $I_A = \frac{ML^2}{3}$

So, $\alpha = \frac{3g}{2L}$

let a be the acceleration of centre of mass,

Then, $a = \frac{L}{2}(\alpha) = \frac{L}{2}\left[\frac{3g}{2L}\right]$

$$a = \frac{3g}{4}$$

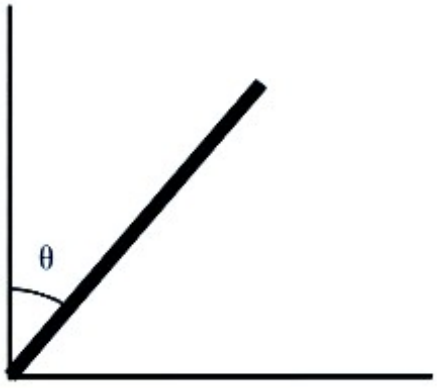
By Newton's second law, for the rod,

$$Mg - N = Ma$$

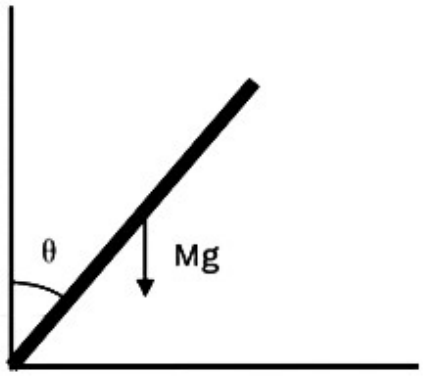
$$N = M[g - a] = M\left[g - \frac{3g}{4}\right]$$

$$N = \frac{Mg}{4}$$

Q. **A thin uniform rod having mass M and length L is hinged at its one end to make it rotate in a vertical plane. Friction is negligible at the hinge. The rod is kept vertical and released. What is the angular acceleration of the rod when it is making an angle θ with vertical?**



Sol.



About the hinge, torque due to Mg will be, $\tau = Mg\left(\frac{L}{2}\sin\theta\right)$

Also, by Newton's second law for rotation, $\tau = I\alpha$

So, $I\alpha = \frac{ML^2}{3}\alpha = Mg\frac{L}{2}\sin\theta$

Angular acceleration of the rod, $\alpha = \frac{3g}{2L}\sin\theta$

Q. **A wheel having a moment of inertia equal to 2.5 kg m² has an initial angular velocity of 40 rad s⁻¹. If a constant torque of 10 Nm is being applied on the wheel, then in what time it accelerates to an angular velocity of 60 rad s⁻¹?**

Sol. Using Newton's 2nd law, $\tau = I\alpha$

$10 = 2.5 \times \alpha, \quad \alpha = 4\,rad/s^2$

By equation of motion in angular motion,

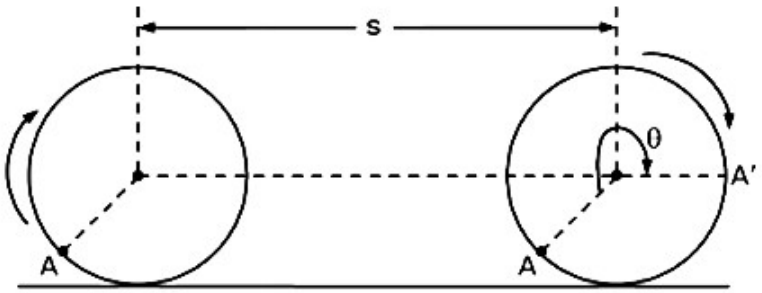
$\omega_f = \omega_i + \alpha t$

$60 = 40 + 4t$

Time taken, t = 5 s

Rolling Motion of Rigid Bodies

Combination of rotation and translation produces pure rolling motion.
When a wheel of a vehicle is moving on a straight path, then its centre of rotation is also moving forward. It can be seen from the motion of wheel that
(a) Centre of wheel is moving forward in pure translation.
(b) A point on the periphery of the wheel follows a complex path which is a combination of translation and rotation about centre of wheel.
A very complex rigid body motion in space can be simplified by considering the motion of symmetric rigid bodies like rings, discs, cylinders, spheres etc.
Let us consider a uniform solid disc having radius R which is rolling without slipping on a horizontal surface.



When the disc rotates by an angle θ, its centre of mass moves a distance, $s = R\theta$.

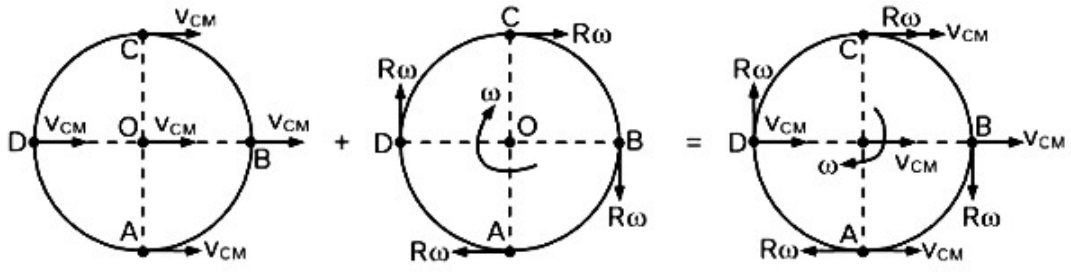
Then, speed of centre of mass, $v_{CM} = \frac{ds}{dt} = R.\frac{d\theta}{dt} = R\omega$

$\Rightarrow v_{cm} = R\omega$

Acceleration of centre of mass, $a_{CM} = \frac{dv_{CM}}{dt} = R.\frac{d\omega}{dt} = R\alpha$

$\Rightarrow a_{CM} = R\alpha$

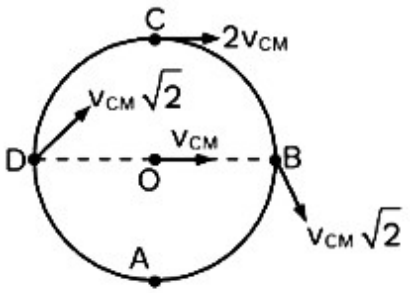
Rolling motion can be expressed as a combination of translational motion and rotational motion i.e., Translation + Rotation = Rolling



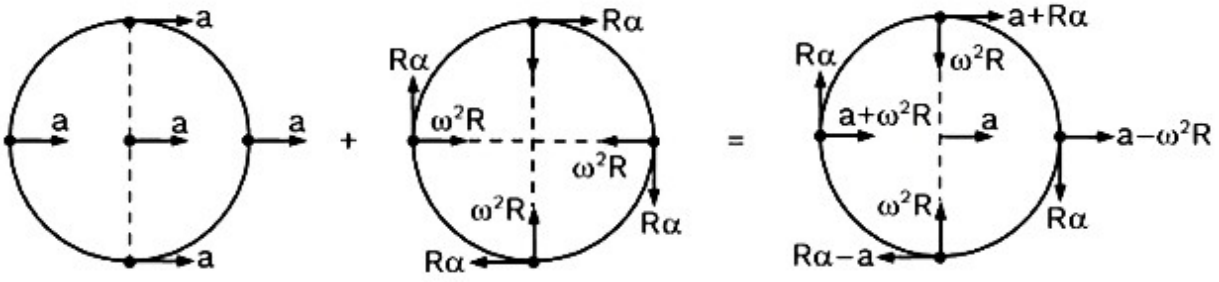
(Pure translation) + (Pure rotation) = (Pure rolling)

If a body is in pure rolling, then net relative velocity at the point of contact should be zero.
i.e., $v_{CM} - R\omega = 0$, $v_{CM} = R\omega$

Thus the lowest point which is in contact with ground has zero velocity. In light of this point, the figure showing various velocities can be redrawn as pure rotation about point of contact A.



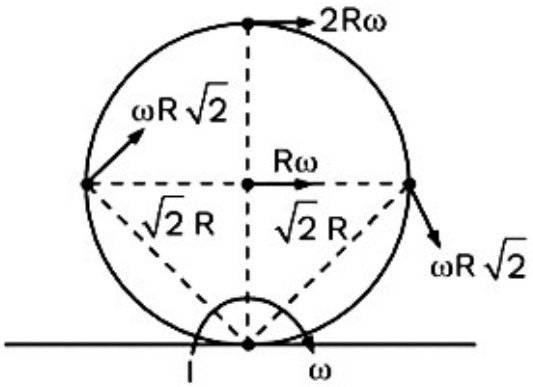
Similarly, acceleration diagram can be drawn,



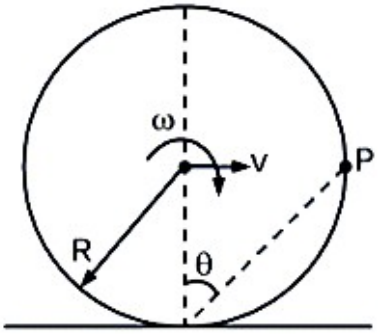
For pure rolling, a = Rα

Rolling as a pure rotation:

Rolling motion of a body can also be viewed as a pure rotation about the point of contact with same angular speed at any instant. The point of contact acts as an instantaneous centre of rotation.



Q. **Find the speed of point P on the surface of disc under pure rolling as shown.**



Sol.

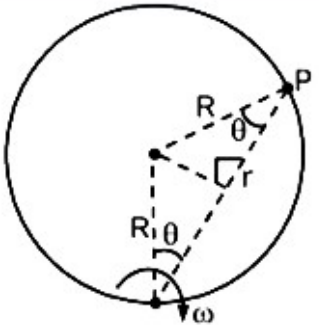
$r = R\cos\theta + R\cos\theta$

$r = 2R\cos\theta$

Now, $v_P = R\omega$

$= 2R\omega\cos\theta$

$v_P = 2v.\cos\theta$



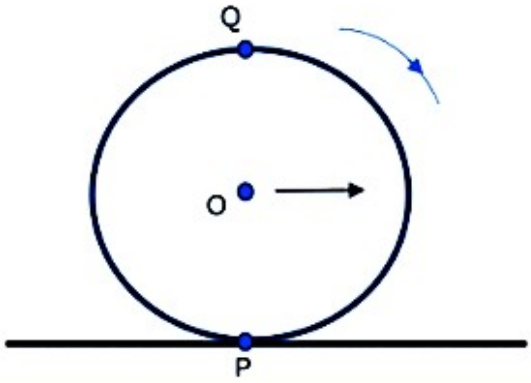
Q. **A sphere is in pure rolling on a fixed horizontal plane surface. P is the point of contact, O is the centre of the sphere and Q is the top most point of the sphere. Then which of the following options are correct?**

a. $\vec{v}_Q - \vec{v}_P = 2(\vec{v}_O - \vec{v}_Q)$

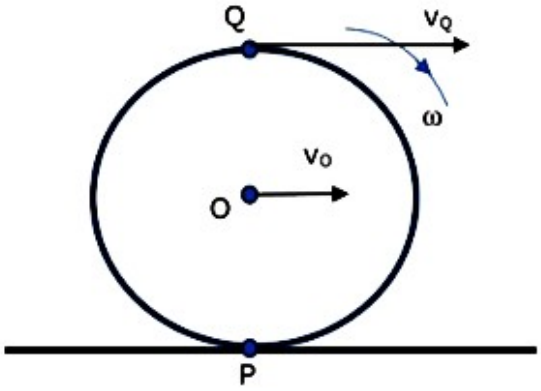
b. $\vec{v}_Q - \vec{v}_C = (\vec{v}_O - \vec{v}_P)$

c. $|\vec{v}_Q - \vec{v}_P| = 2\,|(\vec{v}_O - \vec{v}_Q)|$

d. $|\vec{v}_Q - \vec{v}_P| = 4|\vec{v}_C|$



Sol. If ω is the angular velocity of the sphere and $\vec{v}_P$, $\vec{v}_Q$ and $\vec{v}_0$ are the velocities of the points P, Q and O respectively, then,



$\vec{v}_R = 2\omega R$ about point P

$\vec{v}_C = \omega R$ about point P

$\vec{v}_P = 0$

$\therefore \vec{v}_Q - \vec{v}_C = 2\omega R - \omega R = \omega R$

$\vec{v}_O - \vec{v}_P = \omega R - 0 = \omega R$

$\therefore \vec{v}_Q - \vec{v}_C = \vec{v}_O - \vec{v}_P$ or $|\vec{v}_Q - \vec{v}_P| = |2\omega R - 0| = 2\omega R$

$|\vec{v}_O - \vec{v}_Q| = |\omega R - 2\omega R| = \omega R$

$\therefore |\vec{v}_Q - \vec{v}_P| = 2\,|\vec{v}_O - \vec{v}_Q|$

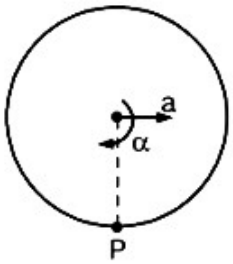
Options b and c are correct.

Accelerated Pure Rolling

In accelerated pure rolling, the linear acceleration of point of contact is zero.

i.e., $a - r\alpha = 0$

$a = r\alpha$



$a_p = a - r\alpha = 0$

$a = r\alpha$

In accelerated pure rolling, if there is friction involved, then it is always static in nature. As the friction is static then it can have any value between 0 to μN. So, while solving the questions of accelerated pure rolling, friction is always treated as an unknown quantity.

Work done by friction is zero in accelerated pure rolling. So, conservation of mechanical energy can be applied for accelerated pure rolling in absence of external and non-conservative forces.

⇒ **Steps for solving problems on accelerated pure rolling**

(a) Choose the system and draw its free-body diagram.

(b) Apply translational equation of motion $\sum F = Ma_{CM}$ for the system ignoring the rotation.

(c) Apply the rotational equation of motion $\sum \tau = I_{CM}\alpha$ for the system ignoring the translation.

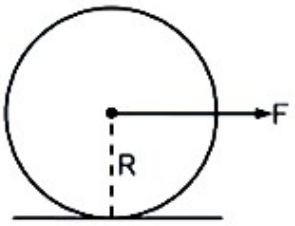
(d) Use the condition for no-slipping $a_{CM} = \alpha R$ or $v_{CM} = R\omega$.

(e) Solve the final equations formed for unknown quantities.

Note: Do not use friction, $f = \mu N$ until it is mentioned that the system is in limiting condition because friction is static in nature in pure rolling motion.

Q. **A force F is applied at centre of a uniform round object of mass m, radius R and moment of inertia about its centre of mass I_{CM}. Find (if $\frac{I_{CM}}{mR^2} = K$),**

(a) Acceleration of centre of the round object if it rolls without slipping
(b) Minimum coefficient of friction required so that the round object rolls without slipping.



Sol. Consider that the friction is f acting opposite to F.

For translational motion,

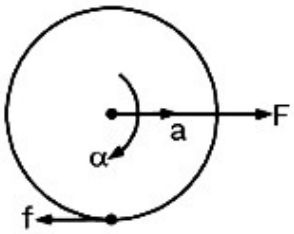
$$F - f = ma$$

For rotational motion,

$$fR = I_{CM} \cdot \alpha$$

$$fR = KmR^2.\alpha$$

$$f = Km(R\alpha)$$



For no slipping,

$$a = R\alpha$$

so, $f = Kma$

then, $F - Kma = ma$

$$F = (K+1)\,ma$$

$$a = \frac{F}{(K+1)m}$$

then, $$f = \frac{KF}{(K+1)}$$

For minimum value of coefficient of friction,

$$\mu_{min}.mg = \frac{KF}{(K+1)}$$

$$\Rightarrow \quad \mu_{min} = \left(\frac{K}{K+1}\right)\frac{F}{mg}$$

Accelerated Pure Rolling on an Inclined Plane:

When a uniform symmetric body is released from rest on a rough inclined plane in such a way that it rolls down without slipping.

For translational motion down the plane,

$Mg\sin\theta - f = Ma$

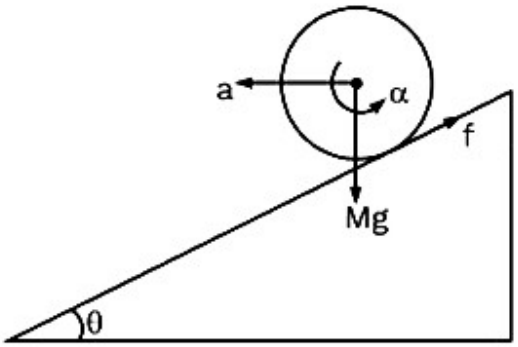
For rotational motion about the centre,

$fR = I_{CM}\,\alpha$

For pure rolling, $a = R\alpha$

so, $fR = I_{CM}\cdot\frac{a}{R}$

$\Rightarrow f = I_{CM}\cdot\frac{a}{R^2}$



$\therefore\ Mg\sin\theta - I_{CM}\cdot\frac{a}{R^2} = Ma$

$Mg\sin\theta = a\left[M + \frac{I_{CM}}{R^2}\right]$

$a = \frac{g\sin\theta}{\left(1+\frac{I_{CM}}{MR^2}\right)}$

For disc, $I_{CM} = \frac{MR^2}{2}$; $a = \frac{2g\sin\theta}{3}$

For solid sphere, $I_{CM} = \frac{2}{5}MR^2$; $a = \frac{5g\sin\theta}{7}$

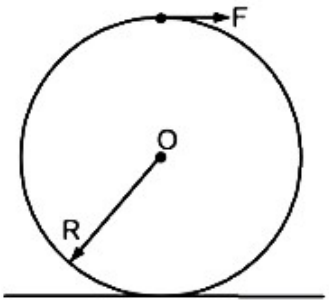
For ring, $I_{CM} = MR^2$; $a = g\frac{\sin\theta}{2}$

For hollow sphere, $I_{CM} = \frac{2}{3}MR^2$; $a = \frac{3g\sin\theta}{5}$

$\Rightarrow a_{\text{solid sphere}} > a_{\text{Disc}} > a_{\text{Hollow sphere}} > a_{\text{Ring}}$

Rolling on Horizontal Surface Without Friction

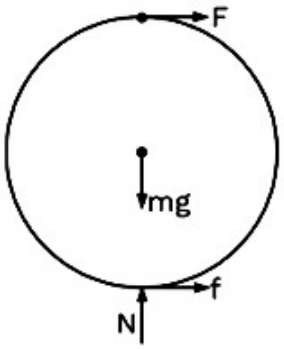
Let's assume that a ring of mass m and radius R is acted upon by a force F as shown in the figure below.



As frictional force is static in nature in rolling motion, so assume it to be acting in forward direction.

By applying Newton's Law,

$$F + f = ma_{CM}$$



For rotation about centre,

$$(F - f)R = I_{CM}\alpha$$

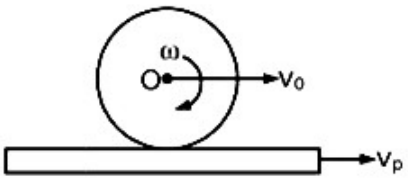
$$(F - f)R = mR^2\alpha$$

$$F - f = ma_{CM}$$

Then, friction, f = 0

So, friction is not required for pure rolling in this case.

Rolling Wheel on a Movable Plank



Lets consider that a round body of radius R is moving on a plank.

Let v_0 be the velocity of the centre of mass of the round body and ω be the angular velocity. Also, the plank is moving forward with a speed v_p as shown in the figure.

By the condition of pure rolling, the points of contact must have same velocity.

thus, $v_0 - R\omega = v_p$

and $a_0 - R\alpha = a_p$ [Here a_0 is acceleration of centre, α is angular acceleration and a_p is acceleration of plank]

For forward slipping

$v_0 - R\omega > v_p$

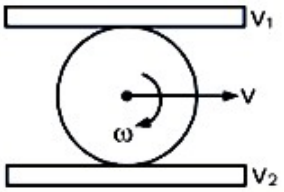
$a_0 - R\alpha > a_p$, when v_0, ω and v_p satisfy the relation $v_0 - R\omega = v_p$.

For backward slipping

$v_0 - R\omega < v_p$

$a_0 - R\alpha < a_p$, when v_0, ω and v_p satisfy the relation $v_0 - R\omega = v_p$.

Q. **A sphere of radius R is rolling without slipping in between two planks as shown in figure. Find**
(a) Speed of centre of sphere
(b) Angular velocity of the sphere.



Sol. For no slipping at upper plank

$$V + R\omega = V_1 \quad \text{...(i)}$$

For no slipping at bottom plank

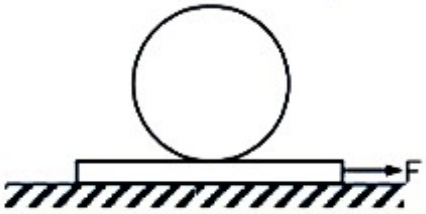
$$V - R\omega = V_2 \quad \text{...(ii)}$$

On adding (i) and (ii), we get

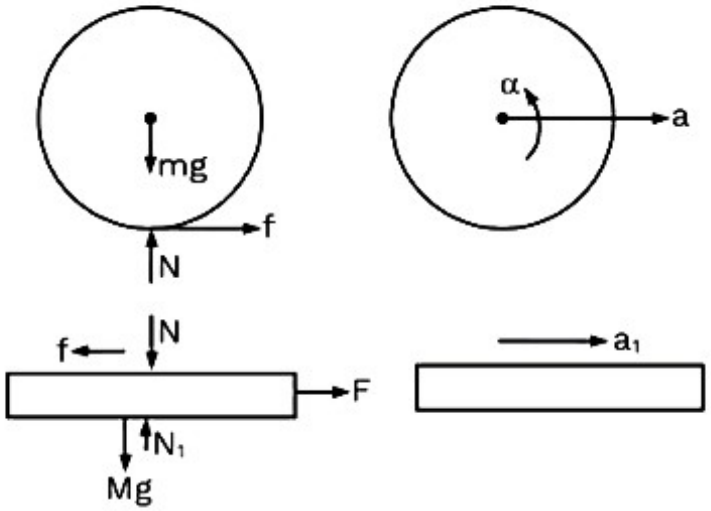
$$V = \frac{V_1 + V_2}{2}, \text{ and}$$

$$\omega = \frac{V_1 - V_2}{2R}$$

Q. **Consider a disc of mass m and radius R placed on a rough plank of mass M which in turn is placed on a smooth horizontal surface. Now plank is pulled by a force F and disc starts to roll on the plank.**
If there is no friction any where then find
(a) Friction force acting on the disc
(b) Angular acceleration of the disc



Sol. Let us apply Newton's Law on disc in horizontal direction



$$f = ma \qquad \text{...(i)}$$

For plank

$$F - f = Ma_1 \qquad \text{...(ii)}$$

Applying torque equation on disc about centre of mass

$$\tau = I\alpha$$

$$\Rightarrow \quad fR = \frac{mR^2}{2}\alpha \qquad \text{...(iii)}$$

For pure rolling

$$a_1 = a + R\alpha \qquad \text{...(iv)}$$

Solving the four equations, we get

$$f = \frac{Fm}{m + 2M}; \alpha = \frac{2F}{(m + 3M)R}$$

Rotational Kinetic Energy

Every particle in a rotating rigid body has same angular speed ω,

So, speed of the ith particle, $v_i = r_i\omega$

The total kinetic energy of the rotating rigid body is the sum of the kinetic energies of the individual particles.

$$K = \sum k_i = \sum \frac{1}{2} m_i v_i^2 = \frac{1}{2} \sum m_i r_i^2 \omega^2$$

$$= \frac{1}{2} \left(\sum m_i r_i^2 \right) \omega^2$$

$$K = \frac{1}{2} I \omega^2, \text{ where } I = \sum m_i r_i^2 = \text{Moment of Inertia}$$

Work Energy Theorem in Rotational Motion:

Rotational work done by a force about a fixed axis is given by,

$$W_{rotational} = \int \vec{\tau} . d\vec{\theta}$$

For translational motion, work – energy theorem is written as,

$$W = \Delta K_{translation}$$

Similarly, Work energy theorem for rotational motion can be written as,

$$W_{rotational} = \Delta K_{rotational}$$

i.e. net rotational work done by forces is equal to change in rotational kinetic energy of the body.

Conservation of Mechanical Energy :

If the work done by forces other than conservative forces is zero, total mechanical energy of a system remains conserved.

i.e.

$$\underbrace{K_i + U_i}_{\text{Intial Mechnical energy}} = \underbrace{K_f + U_f}_{\text{Final Mechanical energy}}$$

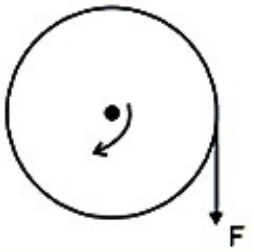
or $\Delta K + \Delta U = 0$

Power :

Rotational power is defined as rate of doing rotational work.

$$P = \frac{d(\vec{\tau} . \vec{\theta})}{dt} = \vec{\tau} . \frac{d\vec{\theta}}{dt} = \vec{\tau} . \vec{\omega}$$

Q. **A uniform cylinder of mass M and radius R, initially at rest is mounted to rotate freely about a horizontal axis that passes through the central axis of the cylinder. A constant force F acts on the cylinder as shown in figure. Calculate the angular velocity of the cylinder as it rotates by an angle of π.**



Sol. Work done by a constant force F is,

$$W = \int \tau.d\theta = \int FR.d\theta = FR\int_0^{\pi} d\theta = FR\pi$$

As per work energy theorem,

$$\frac{1}{2}I\omega_f^2 - \frac{1}{2}I\omega_i^2 = W$$

$$\Rightarrow \quad \frac{1}{2}I\omega_f^2 = FR\pi \text{ ..... (Cylinder was initially at rest)}$$

$$\omega_f = \sqrt{\frac{2FR\pi}{I}} = \sqrt{\frac{4FR\pi}{MR^2}} = \sqrt{\frac{4F\pi}{MR}}$$

Total Kinetic Energy in Rolling Motion :

If a sphere of mass M and radius R is rolling, then,

$$K_{total} = K_{Rotational} + K_{translational}$$

$$= \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}MV_{cm}^2$$

In pure rolling, $V_{cm} = R\omega$

$$\therefore \quad K_{Total} = \frac{1}{2}I_{cm}\omega^2 + \frac{1}{2}M\omega^2R^2$$

$$= \frac{1}{2}\omega^2\left[I_{cm} + MR^2\right]$$

$$= \frac{1}{2}I_p\omega^2 \text{ ..... } [I_p = I_{cm} + MR^2 = \text{ Moment of inertia about point of contact}]$$

Q. **Find the ratio of the rotational kinetic energy to the translational kinetic energy for a solid sphere rolling on a horizontal surface.**

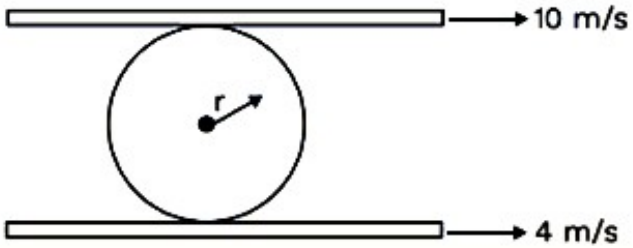
Sol. For a rolling solid sphere, $v = R\omega$

Translational kinetic energy, $K_T = \frac{1}{2}mv^2$

Rotational kinetic energy, $K_R = \frac{1}{2}I\omega^2 = \frac{1}{2}\left(\frac{2}{5}mR^2\right)\omega^2 = \frac{2}{5}\left(\frac{1}{2}mv^2\right)$

So, the ratio of rotational KE and translational KE = 2/5.

Q. **A cylinder of radius r = 10 cm is placed between two planks. If mass of the cylinder is 2 kg, find the kinetic energy of the cylinder assuming that there is no slipping at any point.**



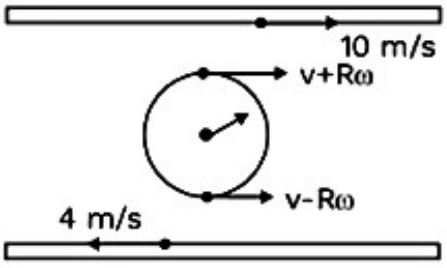
Sol. Let velocity of centre of mass of cylinder be v and ω be its angular velocity.

Velocity of points of contacts should be equal for no slipping.

So, $V + R\omega = 10$

$V - R\omega = 4$

$2v = 14, \quad \boxed{v = 7m/s}$



Then, $R\omega = 3 \Rightarrow \omega = \frac{3}{0.10} = 30$ rad/s

Total kinetic energy of cylinder, $K_E = \frac{1}{2}MV^2 + \frac{1}{2}I_{cm}\omega^2$

$$= \left(\frac{1}{2}\times 2\times 7\right) + \left(\frac{1}{2}\times\frac{MR^2}{2}\omega^2\right)$$

$$= 49 + \left(\frac{1}{2}\times\frac{2\times 0.10^2}{2}\times 30^2\right) = 49 + 4.5 = 53.5\,J$$

Angular Momentum

Angular momentum is rotational analogue of linear momentum.

Angular momentum is a vector quantity.

If a particle of mass m and linear momentum $\vec{P}$ is at a position $\vec{r}$ relative to the origin O, then the angular momentum $\vec{L}$ of the particle wrt the origin O is given by cross product of $\vec{r}$ and $\vec{P}$:

$$\vec{L} = \vec{r} \times \vec{P} \quad \text{... (i)}$$

$$|\vec{L}| = |\vec{r}||\vec{P}| \sin\theta \quad \text{... (ii)}$$

Where $|\vec{P}|$ is the magnitude of $\vec{P}$ and θ is the angle between $\vec{r}$ and $\vec{P}$.

Angular momentum of a particle can be zero if

(a) the linear momentum becomes zero $\left(\vec{P} = 0\right)$, or

(b) the particle is at the origin $\left(\vec{r} = 0\right)$, or

(c) if the angle between $\vec{r}$ and $\vec{P}$ is zero or 180°.

On differentiating equation (i) wrt time, we get

$$\frac{d\vec{L}}{dt} = \frac{d}{dt}\left(\vec{r} \times \vec{P}\right) \quad \text{... (iii)}$$

$$\frac{d}{dt}\left(\vec{r} \times \vec{P}\right) = \frac{d\vec{r}}{dt} \times \vec{P} + \vec{r} \times \frac{d\vec{P}}{dt} \quad \text{... (iv)}$$

Here, $\frac{d\vec{r}}{dt}$ is the velocity of the particle and $\vec{P} = m\vec{v}$.

$\therefore$ $\vec{v} \times m\vec{v} = m\left(\vec{V} \times \vec{V}\right) = 0$ as the vector product of a vector with itself is zero, therefore, equation (iv) becomes

$$\frac{d}{dt}\left(\vec{r} \times \vec{P}\right) = \vec{r} \times \frac{d\vec{P}}{dt}$$

Also, $\frac{d\vec{P}}{dt} = \vec{F}$

So, $\vec{r} \times \frac{d\vec{P}}{dt} = \vec{r} \times \vec{F} = \vec{\tau}$

Hence, $\frac{d}{dt}\left(\vec{r} \times \vec{P}\right) = \vec{\tau}$

and $\frac{d\vec{L}}{dt} = \vec{\tau}$ ... (v)

So the rate of change of the angular momentum of a particle is equal to the torque acting on it.

If the linear momentum and position are given in vector form, then angular momentum can be calculated using the determinant method.

$$\vec{L} = L_x\hat{i} + L_y\hat{j} + L_z\hat{k}$$

$$\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$$

$$\vec{P} = P_x\hat{i} + P_y\hat{j} + P_z\hat{k}$$

Using determinant method, we get

$$(L_x\hat{i} + L_y\hat{j} + L_z\hat{k}) = \begin{vmatrix} \hat{i} & \hat{j} & \hat{k} \\ x & y & z \\ P_x & P_y & P_z \end{vmatrix} = (yP_z - zP_y)\hat{i} + (zP_x - xP_z)\hat{j} + (xP_y - yP_x)\hat{k}$$

Comparing the components of both sides, we get

$$L_x = (yP_z - zP_y) \quad \text{.....(i)}$$

$$L_y = (zP_x - xP_z) \quad \text{.....(ii)}$$

$$L_z = (xP_y - yP_x) \quad \text{.....(iii)}$$

Q. **What is the angular momentum of a car of mass 200 kg travelling on a circular track of radius 50 m with a constant speed of 30 m/s.**

Sol. $\vec{L} = \vec{r} \times \vec{P}$

$|\vec{P}| = mv = 200 \times 30 = 6000$ kg m/s (along the tangent to the circle)

$|\vec{L}| = rP\sin\theta, \theta = 90°$

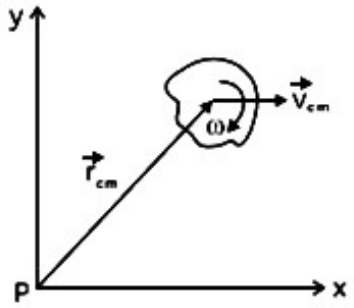
$|\vec{L}| = rP = 50 \times 6000 = 3 \times 10^5 \text{kg m}^2 \text{s}^{-1}$

Angular Momentum for a Mass Rotating About a Fixed Axis

If a particle of mass m is at a distance r from a fixed axis and it is rotating about the fixed axis with an angular speed ω, then its angular momentum about the axis will be,

$L = m(vr) = m(r\omega \times r) = (mr^2)\omega = I\omega$, where I represents the moment of inertia about the fixed axis of rotation.

Angular momentum of a rigid body if the axis of rotation is not fixed



Motion of a rigid body with a translating axis of rotation can be considered to be a combination of pure translation with velocity equal to velocity of centre of mass and pure rotation about centre of mass.

In this case, total angular momentum can be taken as

$$\vec{L}_0 + \vec{L}_{cm} = m\left(\vec{r}_{cm} \times \vec{v}_{cm}\right) + I_{cm}\vec{\omega}$$

As from linear motion, we know that $\vec{F}_{ext} = \frac{d\vec{p}}{dt}$,

Similar, for rotational motion, $\frac{d\vec{L}}{dt} = \vec{\tau}_{ext}$

If the net external torque acting on a system of particles is zero,

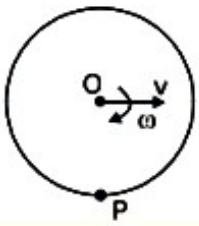
$\Rightarrow \vec{\tau}_{ext} = 0$, then,

$$\frac{d\vec{L}}{dt} = 0$$

Or $\vec{L}$ = constant

Thus, if the total external torque on a system of particles is zero then, the total angular momentum of the system is conserved i.e. remains constant

Q. **A cylinder having a mass of 2 kg and radius 50 cm is moving on a horizontal surface with a linear speed of 25 m/s and its angular speed about the axis passing through the centre of mass is 10 rad/s. Find the angular momentum of the cylinder about point of contact.**



Sol.

$$L_0 = MvR = 2 \times 25 \times \frac{1}{2} = 25 \text{ kg m}^2 \text{ s}^{-1}$$

$$L_{CM} = I_{CM}\omega = \frac{MR^2}{2}\omega = \frac{1}{2} \times 2 \times \left(\frac{1}{2}\right)^2 10 = \frac{10}{4} = 2.5 \text{ kg m}^2 \text{ s}^{-1}$$

$\therefore$ $L_P = L_0 + L_{CM} = 25 + 2.5 = 27.5 \text{ kg m}^2/\text{s}$

Conservation of Angular Momentum

When the external torque acting on a system is zero, then angular momentum of the system remains conserved i.e. $\vec{\tau} = 0$ and

$$\vec{\tau} = \frac{d\vec{L}}{dt}$$

$$\Rightarrow \frac{d\vec{L}}{dt} = 0$$

$$\Rightarrow \vec{L} = \text{constant}$$

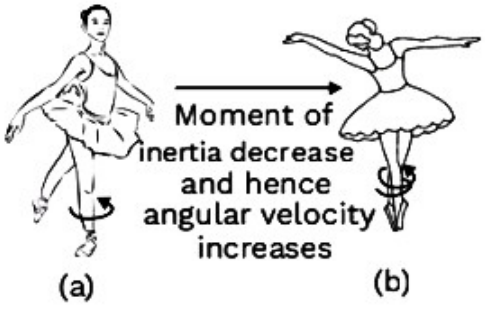
then, $\vec{L} = I\omega = \text{constant}$

Hence, $I_1\omega_1 = I_2\omega_2$

The above equation applies to many situations that we see in our daily lives. Some of them are given below-

(1) On the basis of angular momentum conservation, a ballet dancer performs the dance on her feet. Lets say that a ballet dancer is rotating with her arms and legs stretched outwards. When she folds her arms and brings the stretched leg close to the other leg, her moment of inertia decreases and hence, her angular velocity increases, as shown in the figure given below.

As $\vec{L} = I\vec{\omega}$ = constant and as I decreases, ω increases to keep the angular momentum constant.



(2) An earthen pot maker rotates a disc and keeps mud on it. Then he uses angular momentum conservation to carve out nice pots from the mud.

(3) The angular velocity of revolution of a planet around the sun in an elliptical orbit increases when the planet comes closer to the sun and vice versa. This is because, when the planet comes closer to the sun, its moment of inertia (I) about the sun decreases. As $L = I\omega$ is constant, therefore, its angular velocity ω increases.

Q. **A flywheel which has a moment of inertia 45 kg m² is rotating about its axis with an angular velocity of 10 rad s^{-1}. If the moment of inertia of the flywheel is reduced to 30 kg m² without applying any external torque, then find the new angular velocity of the flywheel.**

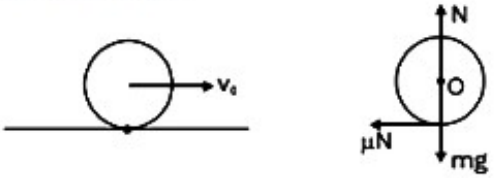
Sol. As no external torque is acting on the flywheel, so, angular momentum remains conserved.

Then, $I_1\omega_1 = I_2\omega_2$

$$\Rightarrow \quad \omega_2 = \frac{I_1\omega_1}{I_2} = \frac{45\times 10}{30} = 15\,\text{rad}\,s^{-1}$$

Q. **A cylinder having a mass m and radius R is placed on a rough horizontal surface after giving its centre a horizontal speed v_0. Find the speed of the centre of the cylinder when it stops slipping.**

Sol. F.B.D. of the cylinder is as follows :



The torque of all the forces about point of contact P is zero.

$$\sum \tau_P = 0$$

$\therefore$ Angular momentum remains conserved about axis passing through point P.

$$L_i = mv_{cm}r_\perp + I_{cm}\omega = mv_0.R + \frac{mR^2}{2}.0 = mv_0R \qquad \text{(clockwise)}$$

$$L_f = mv_{cm}.r_\perp + I_{cm}.\omega$$

$$= m.v.R + \frac{mR^2}{2}\omega \quad \text{(since in pure rolling } v = \omega R\text{)}$$

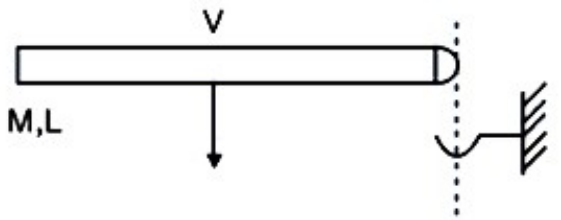
$$= mvR + \frac{mR^2}{2}.\frac{v}{R} = \frac{3}{2}mvR$$

$$L_i = L_f$$

or, $$mv_0R = \frac{3}{2}mvR$$

$\therefore$ $$v = \frac{2v_0}{3}$$

Q. **A rod of mass M and length L is falling vertically with speed v. Suddenly its one end gets stuck in a frictionless hook. Find the angular velocity of the rod just after its end gets stuck.**



Sol. As all the impulse forces during the impact are applied by the hook thus they don't produce any torque about the hook. Thus, angular momentum of the rod about the hook is conserved.

Initial angular momentum about hook $= M\frac{L}{2}v$

Final angular momentum about hook $= \frac{M}{3}L^2\omega$

$$\Rightarrow \quad \frac{ML}{2}v = \frac{ML^2}{3}\omega \Rightarrow \omega = \frac{3}{2}\frac{v}{L}$$

Angular impulse :

Angular impulse is defined as

$$\vec{J} = \int \vec{\tau}_{ext.}\, dt$$

By Newton's second law for rotation,

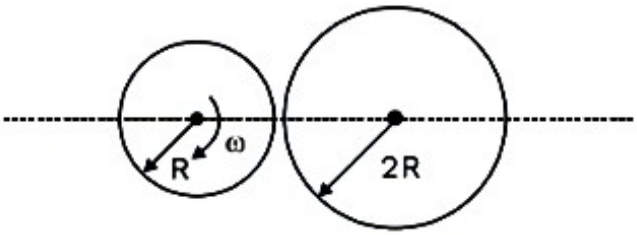
$$\tau_{ext} = \frac{dL}{dt} \Rightarrow \tau_{ext}dt = dL$$

Or, $\int \tau_{ext}dt = \int dL = \Delta L$

$\therefore \quad \vec{J} = \Delta\vec{L} = \vec{L}_f - \vec{L}_i$

Thus, change in angular momentum of a body is equal to angular impulse.

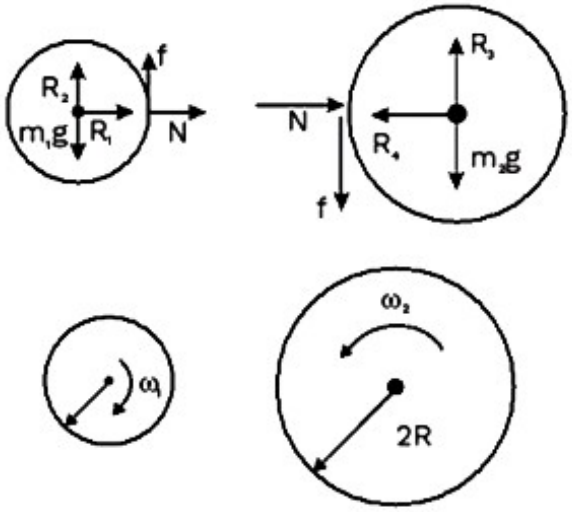
Q. **Two discs of radii R and 2R are pressed against each other. Initially disc with radius R is rotating with angular velocity ω and other disc was stationary. Both discs are hinged at their respective centres and free to rotate about them. Moment of inertia of smaller disc is I and bigger disc is 2I about their respective axis of rotation. Find the angular velocity of the bigger disc after long time.**



Sol. The frictional force causes a torque about the centre of first disc.
Let the discs rotate with angular velocities ω_1 and ω_2 as shown.

Now $\omega_1 R = \omega_2 2R$ as the point in contact should have same velocity after slipping stops.

$$\Rightarrow \quad \omega_2 = \frac{\omega_1}{2}$$



Now, total angular impulse provided by the friction is equal to change in angular momentum of the disc

$$\int fRdt = I(\omega - \omega_1)$$

$$\int f2Rdt = 2I\omega_2$$

$$\Rightarrow \quad 2I(\omega - \omega_1) = 2I\omega_2$$

$$\Rightarrow \quad \omega_1 + \omega_2 = \omega$$

$$\Rightarrow \quad 2\omega_2 + \omega_2 = \omega$$

$$\Rightarrow \quad \omega_2 = \frac{\omega}{3}$$

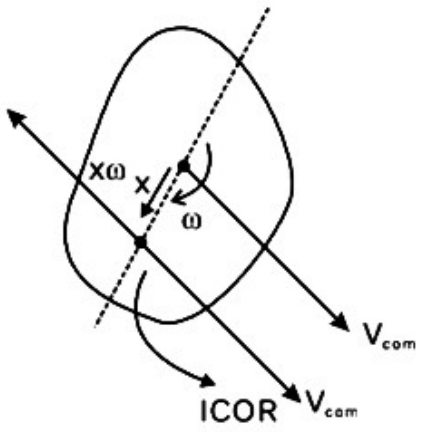
Instantaneous Centre of Rotation

Instantaneous centre of rotation (ICOR) is an imaginary point about which a body can be assumed to be in pure rotation at an instant. This point may or may not be present on the body.

An axis which passes through the instantaneous centre of rotation and is perpendicular to plane of rotation is called instantaneous axis of rotation (IAOR).

Let us consider that an arbitrary shaped rigid body is undergoing translation as well as rotation.

Let the velocity of centre of mass be v_{com} and angular velocity be ω as shown.



The velocity of a point (on the line perpendicular to velocity) is given as

$$\vec{V}_P = \vec{V}_{com} + \vec{\omega} \times \vec{r}$$

If a point P is chosen on this line such that $V_p = 0$, then that point P is called ICOR.

So, we can write,

$$0 = \vec{V}_{com} + \vec{\omega} \times \vec{r} \qquad \text{(For motion in plane } \vec{\omega} \times \vec{r} = |\vec{\omega}||\vec{r}|\sin 90^\circ\text{)}$$

$$r = \frac{|\vec{V}_{com}|}{|\vec{\omega}|}$$

The above equation gives the distance of ICOR from centre of mass of the rigid body.

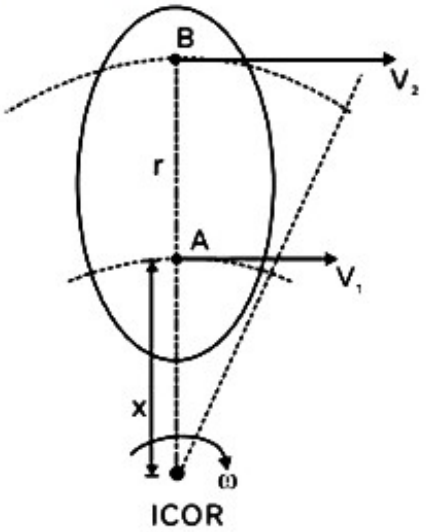
Methods of find ICOR :

The position of ICOR can be located, if the velocities of any two points on the body are known.

(i) For two parallel or antiparallel velocities, the ICOR will lie on the line perpendicular to both the velocities.

Let A and B be two points on an arbitrarily shaped rigid body. Let the points be separated by a distance r, then ICOR will lie on the line joining A and B if their velocities are parallel or antiparallel. Let the ICOR be at a distance x from point A.

Since, $V_{ICOR} = 0$



$\Rightarrow \quad V_1 = x\omega \quad \text{and} \quad V_2 = (x + r)\omega$

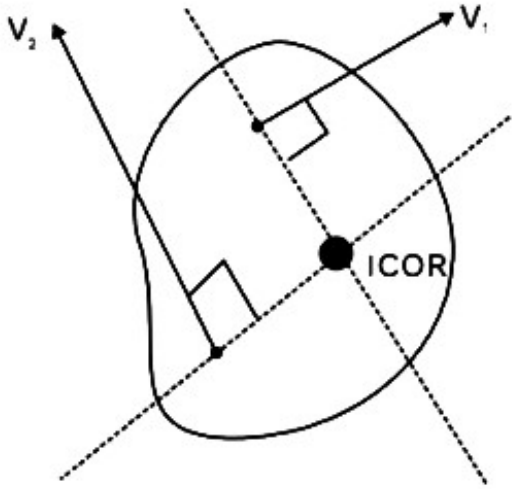
$\Rightarrow \quad \frac{V_1}{V_2} = \frac{x}{x + r}$

$V_1(x + r) = V_2 x$

$V_1 r = (V_2 - V_1)x$

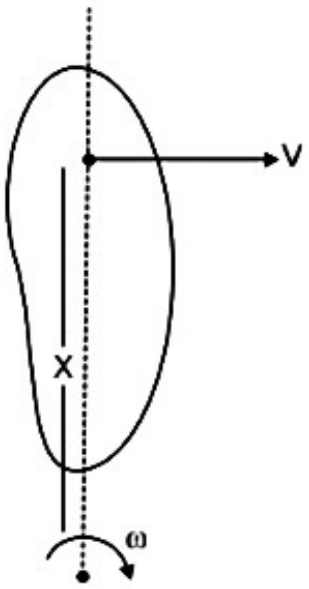
$x = \frac{V_1 r}{V_2 - V_1}$

(ii) When the known velocities of any two points on the rigid body are not parallel, then draw a perpendicular to both the velocity vectors. The point of intersection of the two perpendiculars gives the ICOR.



(iii) If angular velocity and velocity of any point on the rigid body are known, the ICOR will lie on a line perpendicular to the velocity at a distance x. The value of x is given as,

$x = \frac{V}{\omega}$


V
X
ω